

Integrated Audit Decision Log

Provider Services Agreement INS-H1-2024-0417 and the Conditional Reimbursement Agreement referenced in `nkdsjcnkds (3).pdf` and `nkdsjcnkds (2).pdf`.

This integrated audit decision log documents the operational assumptions, identified ambiguities, and standardizing decisions applied during the processing of Provider Services Agreement INS-H1-2024-0417 and the Conditional Reimbursement Agreement referenced in `nkdsjcnkds (3).pdf` and `nkdsjcnkds (2).pdf`.

1. Core Assumptions & Structural Mappings

- **Execution Order (Clause 3.2):** Adjustments are strictly applied in the following order: bundle → plan → premium/uplift → volume discount.
- **Rounding (Clause 3.1):** Integer cents with `half_up` rounding is applied after each individual calculation step, rather than once at the end.
- **Contract Variability:** The Hospital 4 contract structures vary from Hospital 1 (e.g., Base Rates is Section 3, not 4), necessitating extraction by specific clause headers rather than positional alignment.
- **Multipliers & Tiers:** Hospital 1 multipliers for its single facility and plan rate are processed as exactly 1.0×. In Hospital 4, plan tiers are not treated as checkable errors because the contract never defines a closed set of valid tiers.
- **Uplift Omissions:** Non-business-day uplifts are completely absent from the Hospital 4 contract, which is treated as a genuine absence rather than an extraction failure.
- **Exclusion Windows:** Exclusion windows are measured in both directions from the non-billable service's date.

2. Ambiguities Evaluated & Decisions Made

- **Duplicate Billing Scope (Clause 11.3):** The Hospital 4 contract prohibits double-billing the same service, but references a section containing only 18 capped services. This presented two readings: applying the rule only to those 18 capped services (Reading A) or treating it as a universal, free-standing rule (Reading B). **Decision:** Reading B was adopted, flagging cross-invoice and cross-line duplicates for any matched service. For both hospitals, the later occurrence of a duplicate line is flagged and removed from corrected totals.
- **Threshold Premium Base (Clause 5.1):** It was unclear whether premiums apply to all units on a day the threshold is crossed, or exclusively to the specific units exceeding the threshold. **Decision:** The premium is applied to all units once the daily aggregate breaches the threshold.
- **Daily Cap Allocations:** Contracts state per-patient-per-day maximums but omit allocation mechanics across multiple line items. **Decision:** Caps are allocated in ascending `line_id` order, assigning a zero billable quantity to lines exceeding the cap.
- **Volume Discount Scope (Clause 7.1):** “Aggregated across all patients” lacked definition on whether it meant per-patient or hospital-wide. **Decision:** Cumulative utilization is tracked hospital-wide by service date, breaking ties via ascending line identifiers.
- **Reused Invoice IDs:** Contracts do not specify which occurrence of a reused ID is erroneous. **Decision:** Line-item groups are mapped chronologically. The later occurrence is consistently reported as the duplicate across both pipelines.

3. Unresolved Ambiguities & Edge Cases

- **Daily-Cap Allocation Convention:** The literal, ascending line-ID allocation rule fails to reproduce the labelled corrected totals for four invoices in `nkdsjcnkds (3).pdf`. The literal reading is preserved over hard-coding the labels, leaving the intended allocation convention unresolved.
- **Premium Threshold Scope:** The contract text ultimately cannot settle the ambiguity regarding whether premiums should only apply to units strictly above the threshold.
- **Reused ID Policy:** The chronological-mapping rule for reused invoice identifiers remains a calibration choice rather than a strictly contract-derived rule.
- **Price Tie-Breaks with Bundles:** For ambiguous service descriptions (e.g., “CARDIAC physio SESS”), the shared matcher fails to consider bundle-substituted rates as secondary price evidence. Following a strict “flag, don’t guess” policy, these are marked for review rather than resolved.
- **Malformed Dates & Unknown Services:** Invalid dates (e.g., 2025-06-31) and unmatched free-text descriptions are flagged with low confidence, keeping the billed amount as a provisional expectation since a defensible contracted rate cannot be established.